

ALI NIKKHAH

Open to Relocation & Remote | alinkkh9@gmail.com | +98 991 296 3951 | alinikkhah2001.github.io |
linkedin.com/in/alinikkhah2001

SUMMARY

AI/ML Engineer with 2+ years building production agentic systems, multimodal retrieval, and ML infrastructure at scale. At Digikala, led the enterprise AI layer (5M+ daily queries) with LangGraph multi-agent pipelines, hybrid RAG (FAISS + Elasticsearch), and vLLM/Triton serving. At Advanced Analytics, deployed edge CV pipelines on AXIS hardware (TensorRT, Triton, Kafka). At Turquoise, built BNPL credit risk models and financial agentic chatbots with real-time voice transcription. Full-stack ML: PyTorch, distributed training, MLOps, high-concurrency serving.

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led architectural transition to stateful **LangGraph** multi-agent pipelines handling **5M+ daily queries** (p99 <200ms); designed Modular Control Pipeline (MCP) for zero-downtime LLM hot-swapping; built hybrid RAG (FAISS + Elasticsearch via RRF) with HyDE (**+23% recall@10**) and KAG (**-41% hallucinations**).
- Engineered async **FastAPI** serving with vLLM (PagedAttention) and **Triton Inference Server**; multi-tier **Redis** caching and request batching achieving **3.2x throughput, 60% cost reduction per 1K queries**.
- Supervised multi-tenant Kubernetes deployment isolating vision/language model clusters (**200 pods, 40% GPU cost reduction**).

Digikala

Tehran, Iran

AI Engineer

Nov 2024 – Jan 2025

- Prototyped RAG architectures against production logs; integrated LangGraph cyclic graphs for multi-turn sessions; benchmarked retrieval precision/latency to shape R&D roadmap.

Advanced Analytics Australia

Remote Contract

Computer Vision Engineer

Sep 2025 – Apr 2026

- Deployed production video analytics on AXIS edge hardware across **12 sites**: PPE detection, handwash verification, face tracking (**<50ms inference, 99.2% uptime, -35% false positives**); TensorRT/TF Lite via Triton, Kafka for **100+ cameras @ 30 FPS**.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Built BNPL credit risk model (AUC 0.87, **-22% default rate**); engineered LangGraph financial chatbot with Whisper/wav2vec2.0 (**-78% tickets, 94% accuracy**); time-series transformer portfolio allocation (**Sharpe +0.42**).
- Architected data lake (Trino, PeerDB CDC **<60s latency, 5TB/day**); Airflow DAGs, PySpark (**60% memory reduction**); Great Expectations, Prometheus/Grafana observability.

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Automated CI/CD (Docker, ArgoCD, GitHub Actions **<10min releases**); K8s + Terraform; FastAPI/TensorFlow Serving; Ray/MLflow for **15+ models, 99.9% deploy success**.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

RESEARCH HIGHLIGHTS

Univ. of British Columbia (*Remote RA, Feb 2024 – Nov 2025*) — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center (*Remote RA, Apr 2023 – Feb 2024*) — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.